

Design Document: Unbalanced Mach-Zehnder Interferometer

edX UBCx Phot1x, Silicon Photonics Design, Fabrication and Data Analysis | openEBL-2026-10

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Layout file	EBeam_zeyad_874.gds	Top cell	EBeam_Zeyad_v1
Repository fork	github.com/zeyademad202301011/openEBL-2026-10	Date	7 October 2026

1. Objective

This design is a single unbalanced Mach-Zehnder interferometer (MZI) for TE polarization at 1550 nm. The two arms have different lengths, so the transmission spectrum shows periodic fringes. The goal is to measure the free spectral range (FSR) of the fabricated device, extract the group index of the strip waveguide from it, and compare the result with the simulation.

2. Circuit

Component	Compact model	Role in the circuit
Grating couplers (3)	ebeam_gc_te1550	One input for the laser and two outputs for the detectors, placed for the automated fibre-array measurement.
Y-branch	ebeam_y_1550	Splits the input power equally into the two arms.
Strip waveguides	ebeam_wg_integral_1550	The two interferometer arms, with different lengths. Standard PDK strip waveguide (500 nm x 220 nm, oxide cladding).
2x2 broadband directional coupler	ebeam_bdc_te1550	Combines the two arms and gives two complementary outputs.

3. Operating principle

With a Y-branch splitter and a 2x2 coupler as combiner, the two outputs depend only on the phase difference between the arms, $\beta\Delta L$, where $\beta = 2\pi n_{\text{eff}}/\lambda$ and ΔL is the path length difference:

$$T_{1,2} = (1/2) [1 \pm \sin(\beta\Delta L)]$$

The spacing between neighbouring fringes is set by the group index n_g of the waveguide:

$$\text{FSR} = \lambda^2 / (n_g \Delta L)$$

4. Simulation results

Circuit simulation from the layout with Lumerical INTERCONNECT and the SIEPIC EBeam compact model library, 1500 to 1600 nm.

Quantity	Value	Note
FSR near 1550 nm	5.51 nm	Between the dips at 1547.64 nm and 1553.15 nm
Average FSR, 1500 to 1600 nm	5.50 nm	18 dips in the range
Maximum transmission	-5.04 dB / -5.13 dB	Output 1 / output 2, at 1550.4 nm, fibre to fibre
Extinction ratio near 1550 nm	41 dB	Within 5 nm of 1550 nm
3 dB bandwidth of the envelope	43.8 nm	Set by the grating couplers; about 8 fringes inside
Path length difference ΔL	about 104 μm	Derived from the simulated FSR with $\Delta L = \lambda^2 / (n_g \text{FSR})$, using the nominal group index $n_g = 4.2$ of this waveguide. Not read from the layout.

5. Layout and simulated spectrum

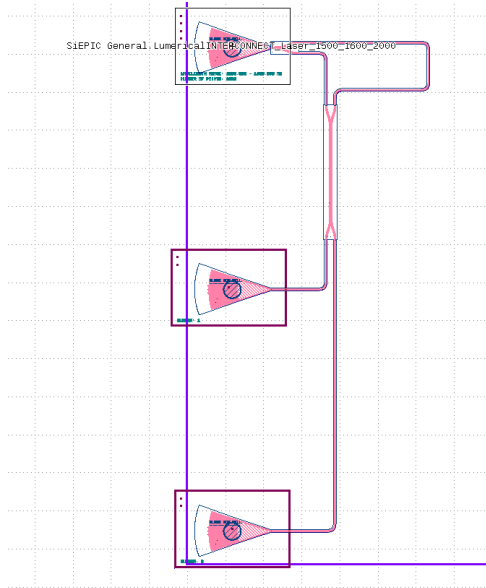


Figure 1. Layout of the MZI in KLayout: three TE grating couplers in a vertical column, with the splitter, the two arms and the 2x2 coupler on the right. The laser and detector markers are simulation symbols and are not fabricated.

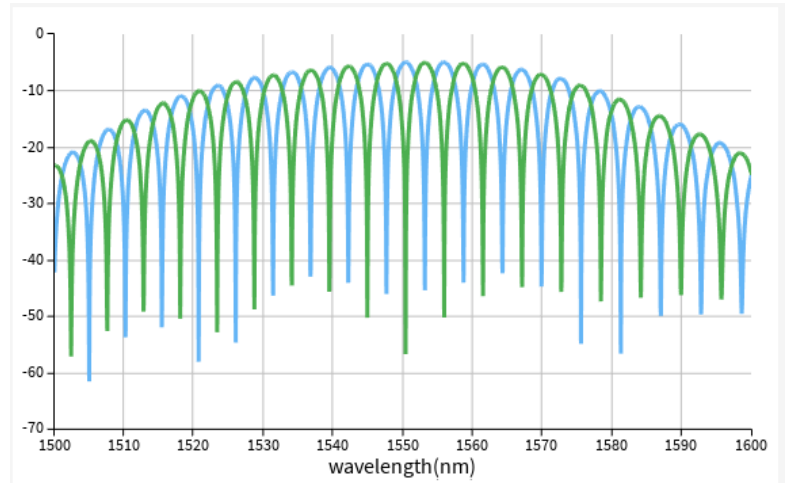


Figure 2. Simulated transmission (dB) of the two outputs from Lumerical INTERCONNECT, 1500 to 1600 nm. The two curves are complementary. The broad envelope is the grating coupler response and the fast fringes are the MZI interference.

6. Analysis plan for the measured data

1. Confirm ΔL from the waveguide lengths in the layout, and use that value in the analysis.
2. Correct the measured spectra for the grating coupler envelope, so that only the MZI fringes remain.
3. Read the FSR from the dip positions and calculate $n_g = \lambda^2 / (\text{FSR} \Delta L)$. Then fit the full MZI transfer function to obtain n_g over the measured wavelength range.
4. Use the two complementary outputs as a consistency check on the fitted values.
5. Compare the measured group index and extinction ratio with the simulated values above, and discuss the differences in terms of the fabricated waveguide width and thickness.